Getting Started with Generative Artificial Intelligence: U-M Instructor Guide

The Getting Started with Generative Artificial Intelligence U-M Instructor Guide provides an overview and resources related to Generative Artificial Intelligence (GenAI or GAI). It provides a quick introduction on GenAI, as well as actionable guidance for teaching in a GenAI era. The information found here is an abridged version of the full content housed on U-M's Generative AI website and is reflective of resources and knowledge available as of September 2023.

Note: This document will be periodically updated when new information and resources are available.

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Introduction to Generative AI

Already familiar with generative AI concepts/tools and its academic implications? Skip ahead to the Five Action Items for Teaching in a Generative AI Era section.



What is Generative AI? What is ChatGPT?

Generative AI (GenAI or GAI) is a general term for artificial intelligence (AI) that creates new content based on their training by neural networks. GenAI is a type of artificial intelligence model or tool which, given a prompt or query, is capable of generating new and creative outputs in a variety of media, including text, images, video, audio, code, or other formats that mimic human creation with varying degrees of success.

ChatGPT is currently the most well known type of GenAI and is a sophisticated chatbot, or conversational artificial intelligence large language model, that has been trained on an enormous collection of text data to develop an understanding of the patterns and structures of human language. U-M recently released its own chatbot, UM-GPT, which is private, secure, and free for U-M faculty and students.

In response to prompts from users, chatbots such as ChatGPT and U-M GPT, can generate text that is coherent and convincingly human-like in seconds. It can summarize historical events, write an essay or basic computer code, translate a passage, or even compose poetry and songs. It is a research assistant, a proofreader, a brainstorming aid, and a calculator. However, outputs fall short of demonstrating the higher levels of learning needed to succeed in a rigorous college setting. Additionally, ChatGPT is also sometimes wrong, but with great confidence.

Introductory Generative AI Resources

- About GenAI
- Generative AI Definitions & Terminology

What are the Academic Implications of Generative AI?

Reports of students using ChatGPT to complete assignments began to spread online shortly after its public release in November 2022. Because large language models can write so quickly and competently, they can assist students in completing many common types of assessments. Text outputs from language models can be used to answer essay questions, write papers, and reply to discussion board posts, simulating recall and understanding with little effort or engagement.

How Quickly can ChatGPT Answer an Essay Question?

Faster than you'd think (video, 3:17 min)

This video reveals just how quickly ChatGPT can answer an open-ended quiz essay question. On average, ChatGPT produces results in a *few seconds* and on rare occasions, it can take up to a few minutes to produce results.

Responsible Use of Generative AI

AI tools are being incorporated into our academic and professional experiences and workflows and are already being integrated into search engines, web browsers, edtech software, spreadsheets, and word processors. As these tools become ubiquitous, it is more productive and of better service to students to focus on helping them surpass AI outputs and to use AI assistance responsibly, rather than prohibiting it altogether. Point students to the ethics of using GenAI for students page for further guidance on the responsible use of GenAI. It is also important to talk with students about ChatGPT and GenAI early and often.

Generative AI Tools



With the rise of numerous GenAI-based tools, it is important to know how to choose a tool that works best for you and your students, if you choose to permit the use of GenAI tools in your course. Consider not only the effectiveness, but the privacy and data collection policies of the tool as you make your choices. Refer to the Artificial Intelligence and U-M Institutional Data information in the Safe Computing Guide for more information.

U-M Developed Generative AI Tools

Artificial intelligence tools developed by the University of Michigan are private, secure, and free for U-M faculty and students. Data you share while using these tools will not be used for training these models, and hence are not at risk of being leaked.

- UM-GPT is U-M's private, secure, and free chatbot for U-M faculty and students.

Newly released tools for advanced GenAI users:

- U-M Maizey is a tool that allows U-M faculty, staff, and students to enrich their GenAI experience based on a custom dataset they provide.
- U-M GPT Toolkit is U-M's most advanced and flexible service offering. It is designed for those who require full control over their AI environments and models. At launch, the API gateway is available upon request.

A more comprehensive listing of *Key In-Depth U-M Generative AI Resources and Support* is available in the Appendix.

Other Generative AI Tools

- The GenAI tools linked below are provided by entities external to U-M. Data you share while using these tools may be used for training models, analytics, etc. and there may be costs associated with using these non-UM tools.
- List of popular AI tools (U-M website)

Five Action Items for Teaching in a Generative AI Era



The primary responsibility of all instructors (faculty and GSIs) is to prepare themselves and their students for GenAI, including: gaining a baseline understanding of what GenAI is, how it's being used, and the benefits and drawbacks to using the technology; best practices for equity-focused, student-centered learning to redesign assignments for anticipated GenAI use; and clearly defining acceptable use of GenAI by students in the classroom. To meet these responsibilities, instructors are encouraged to engage in five action items, which are discussed in the next several pages of this guide:

1. Become familiar with Generative AI
2. Understand the implications Generative AI has on writing
3. Reflect on your course and assessment practices
4. Draft a syllabus statement on Generative AI
5. Talk with students about ChatGPT and GenAI early and often

1. Become Familiar with Generative AI

It is important for all instructors to become familiar with GenAI, regardless of how they decide it will be used or not used in their courses. Watch this informative 3-minute video showcasing a potential application of GenAI, and explore the U-M GenAI website for additional resources and support to familiarize yourself with GenAI and ChatGPT. Also review the GenAI definitions & terminology document and consider attending a workshop or completing a self-paced course.

Generative AI Workshops

Generative AI workshops are being offered across U-M's campus. Visit U-M's Generative AI website and the Generative AI page of Academic Technology@Michigan for listings of workshops and events related to generative AI. A GenAI workshop recordings playlist is also available.

Self-Paced Generative AI Courses

- LinkedIn Learning Generative AI course collection
- Generative AI Teach-Out (Michigan Online)
- ChatGPT Teach-Out (Michigan Online)

Generative AI Prompt Literacy

Regardless of the GenAI tool, the most successful ChatGPT output comes from well-designed prompts. The resources below can help you build your prompt literacy skills.

- Prompt Literacy at a Glance
  This website walks ChatGPT users through how to leverage ChatGPT for academic purposes, ChatGPT prompt best practices, as well as prompt examples.
- Generative AI Prompt Literacy Course
  This is a self-paced course that aims to equip instructors with the essential skill set of forming well-crafted prompts for GenAI tools, such as ChatGPT and U-M GPT.

2. Understand the Implications Generative AI has on Writing

Generative AI-Aware Writing Assignments

Writing is central to U-M's mission to create and publish new contributions to knowledge. Writing is a mode of learning and disciplinary thinking. GenAI technology is developing more rapidly than research can track, and the various instruments present strengths, weaknesses, opportunities, and threats to all aspects of our educational and research mission.

Reference the GenAI In-Depth: Specific Implications for Writing and Other Disciplines page for further impacts of GenAI related to writing and approaches to writing as a template. The page highlights foundation tenets to prepare GenAI aware writing assignments, such as how GenAI will transform traditional academic writing, how GenAI can be used as a tool in writing, and ideas for how to approach the goals of writing assignments differently.

Best Practices for Generative AI Aware Writing Pedagogy

The goal of a writing assignment should be for students to learn from the experience of writing, as opposed to demonstrating what they have already learned. The Sweetland Center for Writing - Teaching Writing with GenAI website explains these practices, how to teach writing with chatbots, course policies for chatbots, citation conventions for GenAI, and provides examples from multiple disciplines. Writing best practices have new implications specific to ChatGPT and perhaps other GenAI environments. Additional GenAI writing best practices can be found in the Writing Instruction in GenAI/GAI Environments document.

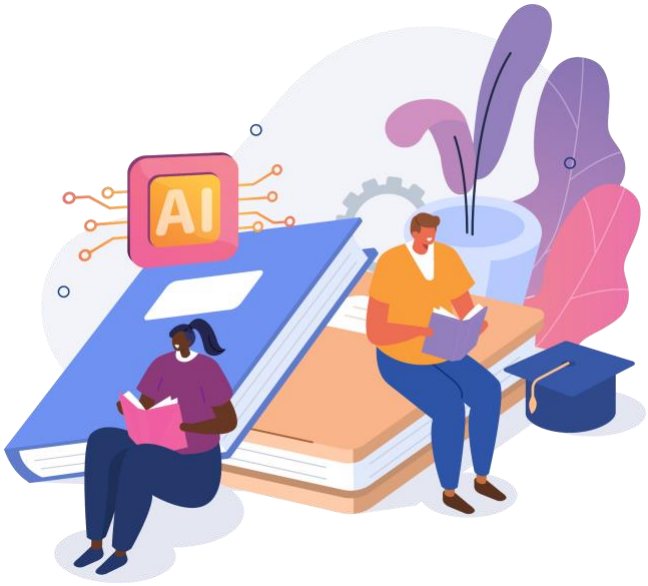
3. Reflect on Your Course & Assessment Practices

Effective pedagogy rises above GenAI's current capabilities. Activities that challenge students beyond recall and superficial understanding exemplify best practice. We encourage instructors to reflect on the following questions about each course:

1. Should GenAI be used in the course or not—and why or why not?
2. If GenAI is to be used, how is the use to be documented?
3. Should course learning objectives be revised?
4. Should GenAI competencies be taught in the specific disciplinary context?
5. Should assessments be revised?

Course Design - Generative AI Implications

We cannot fully anticipate how GenAI will impact every course or assignment at U-M. Educators will need to approach GenAI with an experimental mindset, draw on what we know, prioritize high-impact pedagogical strategies, and plan for incremental, sustainable instructional development. High-impact pedagogy includes collaborative learning, interactive learning, higher-order thinking, service or community-based learning, research, authentic learning and assessment, and writing-intensive courses. That said, writing-intensive courses present particular benefits, opportunities, threats, and risks in a GenAI era.



To begin the process of evaluating course changes, we recommend that instructors practice using GenAI tools relevant to their disciplines, interests, and pedagogies.

In preparation, instructors should evaluate several aspects of their courses:

1. What are the course objectives and rationale for them? Can GenAI be used to meet any of the objectives?
2. Are there new learning objectives in the areas of knowledge, skills, or values about GenAI that students need to meet? Will students have equitable access to GenAI for these objectives?
3. What tasks do students need to complete to demonstrate they meet the learning objectives?
4. How will learning be assessed?
5. Do the objectives, tasks, or assessments present problems with respect to equity, inclusion, diversity, or accessibility? If so, can they be adjusted to ensure fairness and inclusion?

For additional information on course redesign, such as principles of effective pedagogy to guide instructors in adjusting courses to GenAI environments, visit the Course and Assignment (Re)Design page from U-M's Generative AI website, as well as the Designing assignments in the generative AI era document.

Course Assessments - Generative AI Implications

In the age of ChatGPT, true/false, multiple choice, and fill-in-the-blank questions will no longer be adequate means of assessing a student's understanding. In line with best practices for teaching and learning, strategies for engaging and assessing students beyond recall and procedural, superficial understanding are effective pedagogical measures for rising above ChatGPT.

Such assessment approaches include:

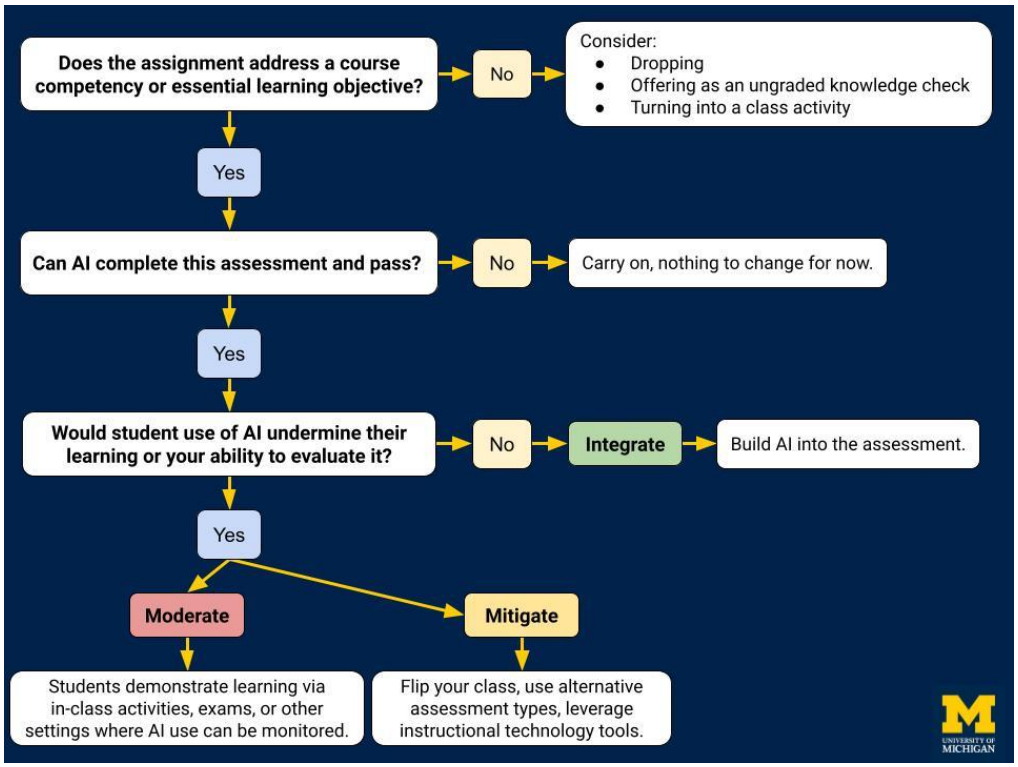
- Writing as a process, with multiple check-ins throughout demonstrating brainstorming, outlining, drafting, and meaningful revision
- Authentic group, individual, and self assessments that require critical and creative thinking
- Applying transparent design principles
- Collaborative and individual presentations

The following chart offers additional assessment approaches for instructors to consider as they redesign courses for GenAI environments:

Pedagogical Strategy	**Possible Assessment Methods**
Active, experiential, and project-based learning sequences similar to those encountered in the discipline	In-class polls, products of peer collaboration, self-assessment
Flipped classrooms to develop interactive learning, peer collaboration, and problem solving skills	In-class polls, products of peer collaboration
Reflections on learning and research processes	Assign a description of research methods, evaluation of sources, explanation of how sources contributed to students' thinking, and discussion of problems encountered and solved in the process Evaluate on the basis of completeness
Live debates, oral presentations, role playing	Evaluation rubric co-created with students before the performances Evaluate on the basis of completeness

Pedagogical Strategy	Possible Assessment Methods
Multimodal compositions, such as infographics, podcasts, drawings, websites, videos, alternative text creation for videos and images	Evaluation rubric co-created with students during scaffolded composition process Authentic group, individual, and self-assessments that require critical and creative thinking
Social annotation of reading	Peer review of annotation quality
Requirements for students to contextualize information, make connections in written work to their lives, current events, and earlier course concepts or lectures, and/or incorporate required readings, videos, images, case studies, or datasets	Evaluate on the basis of completeness and successful integration/synthesis of course materials, concepts, theories, or methods
Poster displays and discussions	Peer review according to a rubric co-created with students during the learning process

Note: Many of the course suggestions in the above chart can be accomplished utilizing the U-M digital tools and Canvas external tools available to instructors. Consultation support is available to instructors for implementing such tools.



Text version of assignment decision tree • Assignment Tree Chart (Google)

Instructors are encouraged to try completing assignments using Generative AI tools before distributing assignments to students. GenAI output can productively be used in class as a prompt for discussion about what technology does well and not so well, as well as the potential social risks and harms implicit in the output (e.g., racial bias, misinformation).

The assignment decision tree, on the previous page, may be used to evaluate whether or not assignments need to be revised while teaching in GenAI environments.

4. Draft a syllabus statement on Generative AI

Teaching in the presence of ubiquitous GenAI tools should be considered a dynamic endeavor and necessitates flexibility. Approaches and expectations will vary by instructor, course, assessment, and activity. To ensure transparency, it is critical that expectations are clearly articulated in the syllabus and continually reinforced when assignments are given. Instructors should align their course policies with their college's academic misconduct policies.

First, familiarize yourself with any revised academic integrity policies from your school/college. Then draft a statement in your syllabus on acceptable uses of GenAI in your classroom.

Course policies might fall into one of three categories:

- Specific uses of GenAI are encouraged (generating ideas, editing, translating, outlining)
- Specific uses of GenAI are allowed if students clearly distinguish between their original work and GenAI output (highlighting output, tracking changes in GenAI output)
- Any use of GenAI constitutes academic misconduct

Read more about course policies and example syllabi statements.

5. Talk with Students about ChatGPT and Generative AI Early and Often

We encourage instructors to set aside time early in the term to tell students directly about course policies regarding GenAI usage, reasons for this decision, and what potential consequences may look like when policies are not followed. We also suggest instructors include language about ChatGPT and GenAI in the class orientation content of Canvas courses. Use the following pointers to hold a conversation with your students about the ethical and practical implications of GenAI for this course, the discipline, and beyond:

- Talk about and evaluate capabilities of GenAI as a group/class
- Motivate students to exceed outputs of GenAI
- Set expectations early, ideally on day one of class, and provide regular reminders
- Policies will vary between campuses, departments, and even individual classes. Each instructor must clearly articulate their own expectations and policies for GenAI use.
- When developing your policies, think about what you actually care about, what you actually want to assess, etc.

- Maximize trust of students to use the tools effectively and ethically, without relying on them entirely
    - Ethics guidance on GenAI for students
- Be explicit about GenAI policies within syllabus language ("In this course…")
- In courses that allow GenAI on some, but not all assignments, be sure to communicate the expectations in each assignment ("For this assignment…")
    - Course Policies & Syllabi Statements
    - Recommended Course Policies, Evals, Strategies to Promote Academic Honesty
- Set and define course policies for disclosure and citation of genAI use
    - MLA Citing Generative AI
    - APA How to Cite ChatGPT
    - Introduction to Academic Integrity (U-M Library page)

Generative AI Ethics, Bias, and Generative AI Detection Tools



Generative AI Ethical Considerations and Bias

GenAI tools, while powerful, should be used with consideration for their bias and limitations. They are trained on large and biased datasets, which can produce misleading outcomes, and have varying degrees of accessibility. As mentioned above, ChatGPT is also sometimes wrong, and with great confidence.

It is also important to know that GenAI models carry implicit biases in them that make them unsuitable for use in cases of ethical deliberation and decision, and should not be used in those circumstances.
Furthermore, ChatGPT is not connected to the internet, and it can occasionally produce incorrect answers. It has limited knowledge of the world and events after 2021 and may also occasionally produce harmful instructions or biased content.

Ethics Resources

- Considering the Ethics of Using GenAI (website)
- Ethics in the Age of Generative AI (LinkedIn Learning course)

Generative AI Detection Tools

AI detection tools, like Turnitin or GPTZero, have become a topic of discussion in the era of GenAI. Instructors should be aware that false positives and negatives are possible and frequent with such detection tools. Because language models generalize and summarize existing knowledge based on probability predictions of word sequences, rather than copying it verbatim, it may be impossible to identify when GenAI has been used with certainty. While AI detection tools may be able to identify the *probability* of AI authorship, they are easily circumvented and cannot provide definitive proof of cheating.

It is recognized that some instructors will want to consider such detection tools to navigate the new and complex landscape of maintaining academic integrity in an AI-infused world. Instructors should be aware, however, that detection tools are imperfect, and currently no software is able to detect AI-generated text with 100% certainty. There is also concern about the potential for detection software to produce so many false positives that it can result in seeding mistrust between students and instructors, compromise access needs, and amplify systemic inequities in higher education. Due to a lack of reliability and transparency in the detection process, such detection tools should not be considered a solution or approach to teaching in a GenAI era. Instead, instructors are encouraged to engage in the five action items for teaching in a GenAI era, outlined in this document.

Appendix: Key In-Depth U-M Generative AI Resources and Support

- U-M Generative Artificial Intelligence (website) for in-depth information on GenAI
  This site offers in-depth explanation of GenAI, guidance on GenAI, and resources such as training, research, AI tools, and prompt literacy.
    - Visit the U-M Guidance for Faculty/Instructors section of the site for GenAI academic/teaching implications



- Generative Artificial Intelligence Committee Report (U-M GenAI Advisory Committee)
  This committee report outlines how U-M can foster responsible, ethical, and equitable use of GenAI in our community and beyond.
    - Reference pages 26-36 of the committee report for near-term guidance for instructors
- Safe Computing guidelines for AI tool use (U-M Safe Computing website)
  This website contains guidance about the use of Artificial Intelligence (AI).
- Generative AI Considerations for Teaching and Learning
  This page from the Academic Technology@Michigan website highlights deeper dive resources on GenAI considerations for teaching and learning.

- Introduction to Academic Integrity (U-M Library page)
  This page from the U-M Library provides an overview of academic integrity, academic ethics, and plagiarism and includes a discussion of ChatGPT. Citation help will be updated often as styles adopt new guides.

- Sweetland Center for Writing - Teaching Writing with GenAI Guide
  This guide, from U-M's Sweetland Center for Writing, outlines why instructors might consider teaching writing with a chatbot, sample writing assignments using chatbots, citation conventions for GenAI and chatbots and information about writing genres and GenAI.

- Course and Assignment (Re-)Design
  This page from the U-M Generative AI website, contains course and assignment (Re-)Design ideas, Strategies for Engagement and Assessment, Integrating AI tools into Course Design, and Skills and Competencies for an AI-Augmented World.
- Designing assignments in the generative AI era
  This guide contains authentic activity examples for faculty/instructors to consider when planning a course in a GenAI era.
- Generative AI Assignment Decision Tree
  (Text version of Generative AI Assignment Decision Tree)

  This decision tree walks instructors through a set of questions to determine whether or not their assignments need revisions when considering teaching and learning in a GenAI era.

Looking for Further Support with Teaching in a Generative AI Era?

U-M has a number of instructional support professionals who can help you understand technologies at U-M, including GenAI, and help you to achieve the best results in your teaching. To request a consultation, fill out the Instructional Support interest form and you will be connected with the most appropriate U-M instructional support group.